Quiz 2

● Graded

Student

Myles Sartor

Total Points

14 / 19 pts

Question 1

Problem 1

6 / 9 pts

- **– 0 pts** Correct
- **– 1 pt** (a) $\binom{5}{2}$ ways to choose 2 men to be on the committee
- **– 1 pt** (a) $\binom{7}{3}$ ways to choose 3 women to be on the committee
- **– 1 pt** (a) $\binom{5}{2}\binom{7}{3}$ ways to choose 2 men and 3 women to be on the committee
- **– 1 pt** (b) $\binom{6}{2}$ ways to choose the rest of 2 women out of remaining 6 women to be on the committee
- **– 1 pt** (b) $\binom{5}{2}\binom{6}{2}$ ways to choose 2 men and 2 women to be on the committee, with one particular women already on the committee
- **– 1 pt** (c) $\binom{3}{2}$ ways to choose 2 men out of the remaining 3 men that can be on the committee
- **– 1 pt** (c) $\binom{3}{2}\binom{7}{3}$ ways to choose 2 men and 3 women to be on the committee, with 2 men cannot be on the committee
- ✔ **– 1 pt** (c) if two particular men cannot be on the committee, the compliment of the event is the union of (the two banned men on the commitee) and (one of the banned men on the committee). There is $\binom{2}{2}$ ways to choose the former event, and there is $2\binom{3}{1}$ ways to choose the latter event. Then the compliment is $1+6=7$ ways.
- ✔ **– 1 pt** the question asked for "number of ways", not probability
- ✔ **– 1 pt** (b) the compliment to the event "one particular women must be on the committee" is the event "one particular women cannot be on the committee", where the number of ways to choose 3 women with one women banned from the committee would be $\binom{6}{3}$. Combining with the ways to choose 3 men, the compliment would be $\binom{5}{2}\binom{6}{3}$.
- **– 0.5 pts** the notations are all inverted, please be careful next time as there would be a greater penalty
- **– 9 pts** your current way of choosing disregarded different genders and lumped sum women and men all together, which is not the question's intention.
- **– 1 pt** In your current way of counting, selecting A then B is different from selecting B then A. In other words, you are counting permutation, but the question asked for combination.
- **– 3 pts** cannot find (c)
- **– 1 pt** should not be addition, should be multiplication between the ways to choose men and the ways to choose women
- **– 3 pts** you should be careful about factorials. The use here does not make sense.
- **– 9 pts** does not make sense

Question 2

Problem 2

4 / 6 pts

- **– 0 pts** Correct answers are:
  a) $\frac{2!5!4!}{9!}$
  b) $\frac{6!4!}{9!}$

- **– 3 pts** major errors in problem

- ✔ **– 2 pts** did not put questions in probability form

- **– 2 pts** part b incorrect

- **– 1 pt** forgot to multiply by 2 in part a

- **– 5 pts** some scratch work, no real work/answer put down

Question 3

Problem 3

4 / 4 pts

- ✔ **– 0 pts** Correct

- **– 0.5 pts** (a) wrong P(B)

- **– 0.5 pts** (a) wrong P(A^B)

- **– 0.5 pts** (a) wrong formula for P(AvB)

- **– 0.5 pts** (a) wrong answer

- **– 2 pts** (a) wrong

- **– 0.5 pts** (b) wrong formula

- **– 0.5 pts** (b) wrong P(B)

- **– 0.5 pts** (b) wrong P(A^B)

- **– 0.5 pts** (b) wrong answer

- **– 2 pts** (b) wrong

Quiz 2

Name: Myles Sartor Student ID: 119017708

Problem 1. Out of 5 men and ~~7~~ 6 women, a committee consisting of 2 men and 3 women is to be formed. In how many ways can this be done if

12

5

(a) any men and any women can be included?

(b) one particular women must be on the committee?

(c) two particular men cannot be on the committee?

a) $$\frac{\binom{5}{2}\binom{7}{3}}{\binom{12}{5}}$$

P(all men) = $$\frac{\binom{5}{2}}{\binom{12}{5}}$$

b) $$1 - \frac{\binom{5}{2}\binom{7}{3}}{\binom{12}{5}}$$

c) $$\frac{\binom{5}{2}\binom{7}{3} - \binom{7}{3}\binom{2}{2}}{\binom{12}{5}}$$

Quiz 2

Name: Myles Sartor Student ID: 119017708

Problem 2. Assume we have 4 boys and 5 girls. We want to arrange them to sit in a row.

a. What is the probability that the boys and the girls are each to sit together? B G

b. What is the probability if only the boys must sit together?

Total: 9!

a) $4!\,5!\,2!$

5 girls

2 groups/subjects

4 boys

$\underbrace{b_1 b_2 b_3 b_4}_{1}\ \underbrace{g_1 g_2 g_3 g_4}_{1}$

2 units/blocks

$g_1 g_2 g_3 b_1 b_2 b_3 b_4 g_4 g_5$

$g_1 b_1 g_2 b_2 g_3 b_3 g_4 b_4 g_5$

$g_1 g_2 B g_5 g_3 g_4$

$\underbrace{g_1 g_4}_{1}\ \underbrace{B}_{1}\ \underbrace{g_3 g_2 g_5}_{1}$

b) $3!\,4!\,5!$

4 boys

5 girls

3 groups/subjects

Quiz 2

Name: Myles Sartor Student ID: 119017708

Problem 3. Suppose $P(A) = 0.4$, $P(B') = 0.3$, and $P(A \cap B') = 0.1$.
(a) Find $P(A \cup B)$.
(b) Find $P(B \cap A')$.

$0.8 - 0.7 = 0.1$

$P(A \cup B) - P(B) = P(A \cap B')$

$0.3 + 0.1 = 0.4$

$P(A \cap B) + P(A \cap B') = P(A)$

$P(A \cap B) = P(A) - P(A \cap B')$

$= 0.4 - 0.1 = 0.3$

$P(B \cap A') + P(B \cap A) = P(B)$

$0.4 + 0.3 = 0.7$ ✓

$1 - 0.3 = P(B)$

$= 0.7$

a) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$0.4 + 0.7 - 0.3$

$= 0.8$

b) $P(B \cap A') =$

$P(B \cap A') = P(B) - P(B \cap A)$

$= 0.7 - 0.3$

$= 0.4$